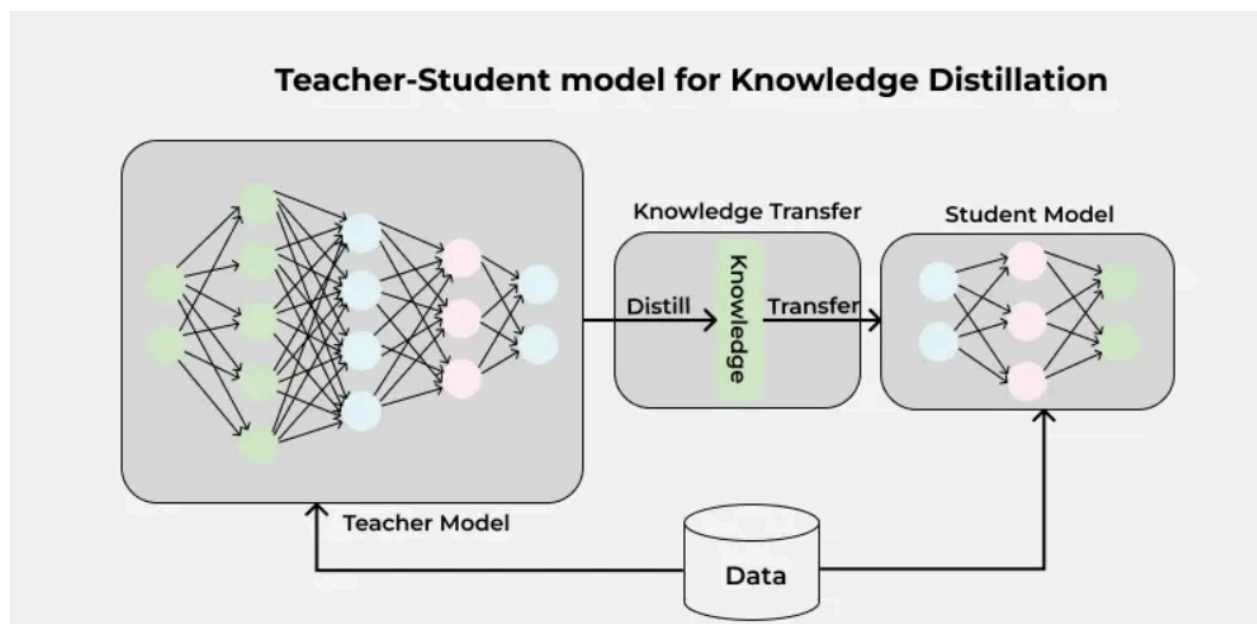


DistilBERT Model

DistilBERT is a distilled version of BERT meaning it is trained using **knowledge distillation** a technique where a smaller model (student) learns from a larger model (teacher). It retains 97% of BERT's performance while being 40% smaller and 60% faster making it highly efficient for NLP tasks such as text classification, sentiment analysis and question-answering

How DistilBERT Works?

DistilBERT utilizes **knowledge distillation** where a smaller model (student) learns to replicate the behavior of a larger model (teacher). This process involves training the student model to mimic the predictions and internal representations of the teacher model.



In the above diagram the **teacher model** (BERT) is a large neural network with many parameters. The **student model** (DistilBERT) is a smaller network trained to replicate the teacher's behavior using **knowledge transfer**. The **distillation process** involves minimizing the difference between the teacher's soft predictions and the student's output allowing the student model to retain most of the teacher's knowledge while being significantly smaller.

Training DistilBERT

DistilBERT is trained using a **triple loss function** which combines:

1. **Language Modeling Loss** – Predicts the next word in a sentence.
2. **Distillation Loss** – Encourages the student model to mimic the teacher's soft predictions.
3. **Cosine-Distance Loss** – Aligns the hidden state representations of the student and teacher models.

By combining these losses DistilBERT is able to learn efficiently from BERT while maintaining high

Limitations of DistilBERT

While DistilBERT is impressive, it's not without trade-offs. The reduction in size means it may struggle with extremely complex language tasks where BERT's deeper architecture excels. For cutting-edge research or niche applications requiring peak performance, the original BERT or even larger models like RoBERTa might still be preferred.

DistilBERT offers an excellent balance between performance and efficiency, making it a go-to choice for many NLP applications. Whether you're working on sentiment analysis, question answering, or any other NLP task DistilBERT is a powerful tool that can help you achieve great results without breaking the bank on computational resources.

Feature	BERT-base	DistilBERT
Layers	12	6
Hidden size	768	768
Attention heads	12	12

Parameters	110M	~66M
Size reduction	—	~40% smaller
Speed	—	~60% faster

ALBERT Model

Factorization of the Embedding Matrix

In BERT-based models, the input embeddings and hidden layer embeddings have the same size, which increases the number of parameters.

In this model, the embedding matrix is split into two parts. The input embedding is smaller and is used to learn word meaning without context. The hidden embedding is larger and is used to learn contextual information. A projection layer connects the two embeddings.

This approach reduces the number of parameters by about 80% with only a small drop in performance.

Cross-Layer Parameter Sharing

In models like BERT, each encoder layer has its own parameters, even though many layers learn similar patterns.

To reduce redundancy, this model shares parameters across layers. This makes the model more efficient and smaller in size.

The authors propose three ways to share parameters:

1. Share only the feed-forward network parameters.
2. Share only the attention parameters.
3. Share all parameters across layers (used as the default setting)

Inter-Sentence Coherence Prediction

Similar to BERT, ALBERT uses Masked Language Modeling during training. However, instead of using the Next Sentence Prediction (NSP) task, ALBERT introduces a new objective called Sentence Order Prediction (SOP).

NSP is a binary classification task that predicts whether one sentence follows another in the original text. A limitation of NSP is that the model can rely on topic similarity rather than learning true sentence coherence.

To overcome this issue, SOP uses pairs of sentences taken from the same context and trains the model to predict whether their order is correct or swapped. Since both sentences belong to the same topic, the model must focus on understanding sentence coherence and logical flow